

## Assignment -1

## Artificial Intelligence – C461

### Instructions:

- You can do it individually or in a group of 2.
- This is a hand-written assignment. Submit scanned copy on gcr (Submit code as .py file).
- Write python code for Question 1 and 3.
- Deadline is 18<sup>th</sup>, January 2026 5:30 PM.
- Any plagiarism will lead to negative marking for both parties.
- Clearly mention assumptions wherever required.

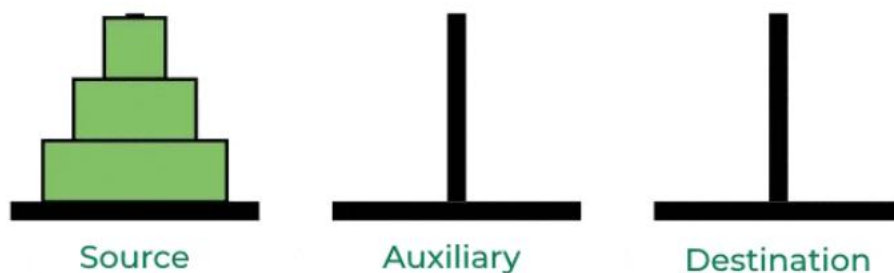
### Question 1: Tower of Hanoi

(Marks = 30)

**Tower of Hanoi** is famous mathematics puzzle with 3 rods and 3 disks. First, all the disks are arranged in first rod from larger disk to smaller disk. We have to move all the disk from the first rod to last rod in same arrangement from larger to smaller.

There are following rules in Tower of Hanoi:

- Only one disk can be moved at one point of time.
- Smaller disk can be placed above larger disk, but larger disk cannot be placed above smaller disk.

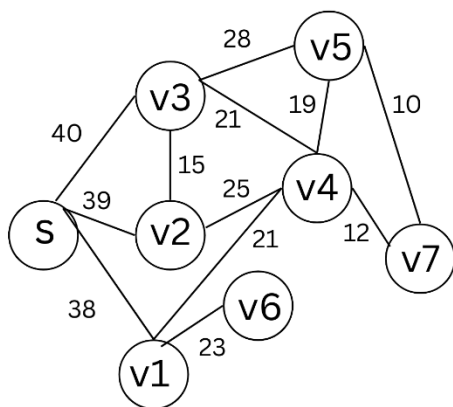


Apply BFS and DFS to achieve final goal when initial state is given. Show all the steps particularly configuration of close and open lists.

**Question 2: Route Finding Problem**

(Marks = 30)

Consider the following map:

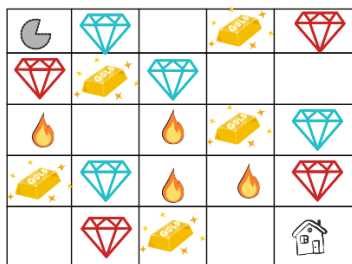


Initial state is s and goal state is v7. Perform a comparative analysis between DFS and UCS to find which algorithm does best in terms of cost (km). Show all the steps.

**Question 2: Maximize Pacman profit**

(Marks = 40)

A pacman board is given:



Initial position of Pacman agent is (0,0) while goal state is (4,4). Apply BFS and IDS (Iterative Deepening Search) such as the profit is maximized. Pacman agent must maximize his profit by collecting treasure. Each treasure has different values, which are given below:

Gold= 120 points

Green Diamond= 80 points

Red Diamond= 110 points

Fire= -50 points

Show all the steps. Also write python code for given problem.